

QUEUE USING LINKEDLIST

DATE:27-10-2025

CODE:

```
#include <stdio.h>

#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

struct Node *front = NULL, *rear = NULL;

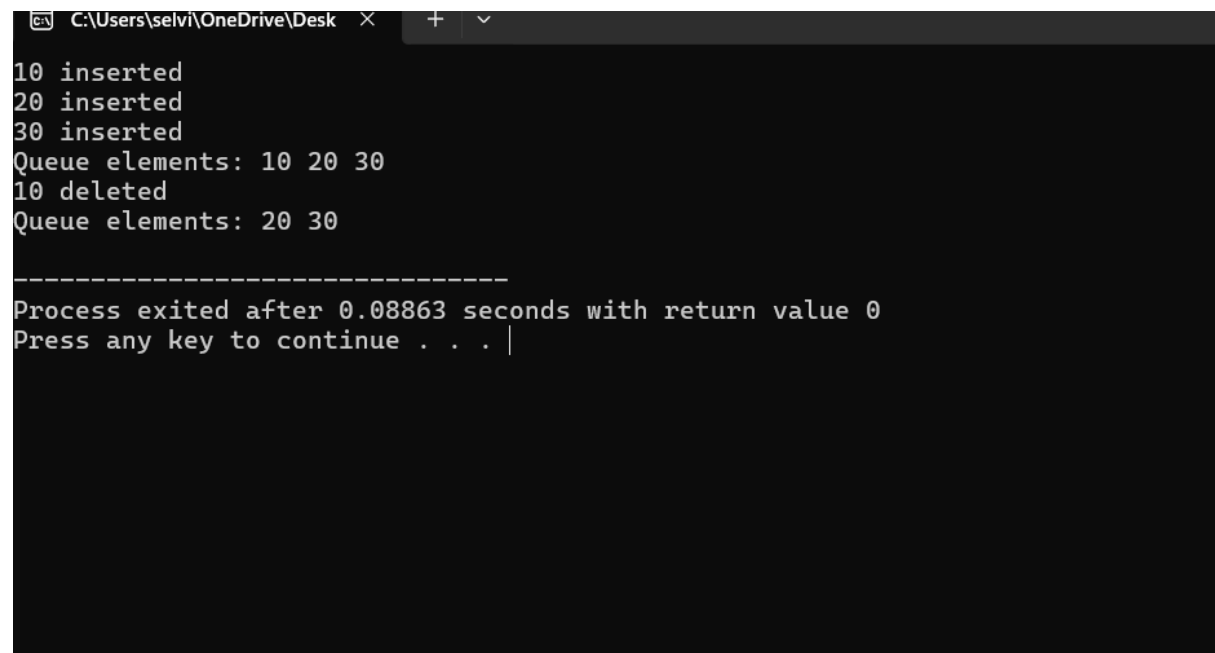
void enqueue(int value) {
    struct Node *newNode = malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;
    if (rear == NULL)
        front = rear = newNode;
    else {
        rear->next = newNode;
        rear = newNode;
    }
    printf("%d inserted\n", value);
}
```

```
void dequeue() {  
    if (front == NULL)  
        printf("Queue is Empty\n");  
    else {  
        struct Node *temp = front;  
        printf("%d deleted\n", front->data);  
        front = front->next;  
        if (front == NULL) rear = NULL;  
        free(temp);  
    }  
}
```

```
void display() {  
    struct Node *temp = front;  
    if (temp == NULL)  
        printf("Queue is Empty\n");  
    else {  
        printf("Queue elements: ");  
        while (temp) {  
            printf("%d ", temp->data);  
            temp = temp->next;  
        }  
        printf("\n");  
    }  
}
```

```
int main() {  
    enqueue(10);  
    enqueue(20);  
    enqueue(30);  
    display();  
    dequeue();  
    display();  
    return 0;  
}
```

OUTPUT:



```
C:\Users\sarvi\OneDrive\Desktop  
10 inserted  
20 inserted  
30 inserted  
Queue elements: 10 20 30  
10 deleted  
Queue elements: 20 30  
-----  
Process exited after 0.08863 seconds with return value 0  
Press any key to continue . . . |
```